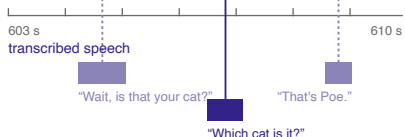


A

BabyView 2026.1 · 51 children · 16,306 recordings
frames, 1 per second



pair each utterance with the frame at its midpoint second
no score selects the frame

1,838,288 utterances · 1,745,489 frames on disk

1,686,105 training pairs

after English filter: -1,258 videos, 48 children

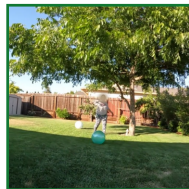


B

Gemini reads each pair: alignment 0–100 + referent noun



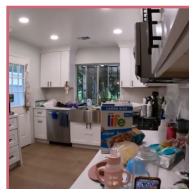
"Blinker, blinker, little car."
aligned 100 · referent: car



"Come to Bobo, I mean, ball."
aligned 100 · referent: ball



"Pria, try your fork."
aligned 0 · no referent



"Let's call him later."
aligned 0 · no referent

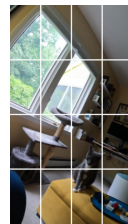
1,686,105 all pairs (100%)

171,793 about something visible (10%)

110,577 ...and the referent is spoken (7%)

C

frame



frozen encoder
whole-image (CLS) token
+ 4x4 region grid

utterance

"Which cat is it?"



bag of words, learned
from scratch

word · region
similarity
score = max
over regions



evaluation: "cat"? 60-way 4AFC, out-of-corpus photos



cat



ball



dog



hat

chance 25%